Business Finance Midterm Cheat Sheet: Ch.2–7

Formulas · BA II Plus keystrokes · Key concepts | Prof. Gemma Lee, Kyung Hee Univ. | Lecture-slide scope only

0. BA II Plus Setup

0.1 Reset (before every problem)

- `2nd CLR TVM` clears N, I/Y, PV, PMT, FV (CE|C, ON/OFF, 2nd QUIT do **not**)
- Cash-flow (NPV) problems: `2nd CLR WORK`
- P/Y = 1: `2nd P/Y` `1 ENTER` `↓` `1 ENTER` `2nd QUIT`
- 4 decimals: `2nd FORMAT` `4 ENTER` `2nd QUIT`
- BGN on screen → switch to END (coupons and ordinary annuities pay at period end)
- Full reset (`2nd RESET ENTER`) sets P/Y back to 12 → reset P/Y = 1

0.2 Input rules

- Enter 4 of N, I/Y, PV, PMT, FV; `CPT` + the 5th
- Cash out = **−**, cash in = **+**; same sign on PV and FV → Error 5
- I/Y = rate per period as a % number (1% → 1)
- Single sum → PMT = 0
- Negative answer = amount to pay now, not an error
- Answer off by exactly (1 + r) → BGN mode is on

Ch.2 Financial Statements, Taxes and Cash Flow

2.1 Balance Sheet

- Snapshot at a point in time; **Assets = Liabilities + Equity**
- Assets in decreasing liquidity; liabilities by when due
- NWC = current assets − current liabilities
- Liquidity = speed and ease of conversion to cash; avoids financial distress

2.2 Book vs. Market Value; Equity Accounts

- Book = accounting value (historical cost); market = price in actual trade → use market value for decisions
- Common stock = shares × par; paid-in surplus = (issue price − par) × shares
- Equity = paid-in capital + retained earnings; par = arbitrary nominal amount
- Ex: 100 sh, par $1, issued at $6 → $100 + $500 = **$600**

2.3 Income Statement

- Performance over a period; sales first, then subtract costs
- Sales − COGS − Depreciation = **EBIT**; EBIT − Interest = taxable income; − Taxes = **NI**
- NI = dividends + addition to retained earnings
- Ex: 1,509 − 750 − 89 = 670; 670 − 70 = 600; tax 21% = 126; NI = **474**
- Trap: interest is deducted **after** EBIT

2.4 GAAP vs. IFRS

- GAAP: FASB, rules-based | IFRS: IASB, principles-based | Korea: K-IFRS
- Same purpose; presentation order flipped (GAAP current items first, IFRS non-current first)

2.5 Taxes

- Average = total tax ÷ taxable income | Marginal = rate on the next $1 (use for decisions)
- Flat tax → marginal = average; progressive → marginal > average
- US flat 21% (2017 reform; before: progressive up to 35%) | Korea progressive 10/20/22/25%
- OECD Pillar Two: 15% minimum effective tax for large MNEs (≥ €750M)
- Ex: Korea ₩500M: 10% × 200M + 20% × 300M = ₩80M → average **16%**, marginal **20%**

2.6 Cash Flow (CFFA)

- **CFFA = OCF − NCS − ΔNWC = CF to creditors + CF to stockholders**
- OCF = EBIT + Depreciation − Taxes

- NCS = End net fixed assets − Beg net fixed assets + Depreciation
- ΔNWC = End NWC − Beg NWC
- CF/CR = Interest − Net new borrowing | CF/SH = Dividends − Net new equity
- Ex: OCF = 670 + 89 − 126 = 633 | NCS = 1,709 − 1,644 + 89 = 154 | ΔNWC = (1,403 − 389) − (1,112 − 428) = 330
- Ex: CFFA = 633 − 154 − 330 = **149** | check: CF/CR 70 − 46 = 24, CF/SH 165 − 40 = 125, sum 149
- Ex (Zephyr): 3,180 − 1,400 − 600 = **1,180**
- Traps: depreciation is added back (OCF, NCS) | current-liability increase lowers NWC | negative CF/CR = borrowing > interest paid | omit any component → wrong
- Cash flow ≠ accounting statement of cash flows

Ch.3 Working with Financial Statements

3.1 Common-Size Statements; Users

- Balance sheet ÷ total assets | income statement ÷ sales; compare across years and across different-size firms
- Ex: inventory 84,600 ÷ 470,000 = **18.0%**
- Internal users: evaluation, planning, control | External: lenders, suppliers, customers, investors, competitors / acquirers / regulators

3.2 Ratio Categories

- Liquidity (short-term solvency) | Leverage (long-term solvency) | Turnover (asset management) | Profitability | Market value
- Slide firm: Sales 2,361; NI 474; TA 3,630; TE 2,625; price $115; 33M shares

3.3 Liquidity

- Current = CA ÷ CL → **1.29**
- Quick = (CA − Inventory) ÷ CL → **0.52** (excludes inventory)
- Cash = Cash ÷ CL → **0.18**

3.4 Leverage and Coverage

- Total debt ratio = (TA − TE) ÷ TA → **0.28**
- D/E = TD ÷ TE → **0.38** | EM = TA ÷ TE = 1 + D/E → **1.38**
- TIE = EBIT ÷ Interest = 741 ÷ 141 → **5.26**
- Cash coverage = (EBIT + Dep.) ÷ Interest = 1,017 ÷ 141 → **7.21** (always ≥ TIE)

3.5 Turnover

- Inventory = **COGS** ÷ Inventory → 3.18 (numerator is COGS, not sales)
- Receivables = Sales ÷ AR → 12.56 | Payables = COGS ÷ AP → 3.91
- Total asset turnover (TAT) = Sales ÷ TA → 0.65

3.6 Profitability

- PM = NI ÷ Sales → 20.08% | ROA = NI ÷ TA → 13.06% | ROE = NI ÷ TE → 18.06%
- Same numerator; only the denominators differ

3.7 Market Value

- EPS = NI ÷ shares → 14.36 | P/E = price ÷ EPS → 8.01
- P/S = price ÷ (Sales/share) → 1.61 | M/B = price ÷ (TE/share) → 1.45 (book value per share)
- EV = market cap + debt − cash; EBITDA = EBIT + Dep.
- Ex: EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 1,017; **EV/EBITDA = 4.62**

3.8 DuPont

- **ROE = PM × TAT × EM** (a product, not a sum) | PM = profitability / cost control, TAT = asset efficiency, EM = leverage
- Ex: Coca-Cola 27.34% × 0.46 × 3.26 ≈ **40.7%** (slide value; inputs rounded)
- KO vs PEP: both ROE ≈ 40%, different paths
  - KO: sells concentrate, asset-light, high margin, low turnover
  - PEP: finished goods and snacks, ~2× sales, thin margin, high turnover and leverage
  - PEP current ratio < 1 is not a red flag (cash sales, long payables terms, strong credit)

3.9 Benchmarks and Limits

- Benchmarks: time-trend analysis; peer group (SIC / NAICS)
- Limits: no underlying theory, conglomerates, global (IFRS vs GAAP), accounting methods (FIFO / LIFO, depreciation), different year-ends, capital structure, seasonality / one-offs, window dressing

Ch.4 Time Value of Money

4.1 Concepts

- A dollar today > a dollar tomorrow (it can be invested)
- Compounding (interest on interest) ↔ discounting
- Simple = interest on principal only | compound = on principal + past interest; gap grows with time
- Ex: $1,000, 10%, 3 yrs: simple 1,000 + 3 × 100 = 1,300 vs compound 1,000 × 1.10^3 = **1,331** (interest-on-interest $31)

4.2 Formulas

- $FV = PV(1 + r)^t$ | $PV = FV \div (1 + r)^t$
- $r = (FV \div PV)^{1/t} - 1$ | $t = \ln(FV \div PV) \div \ln(1 + r)$
- Slides favor `CPT I/Y` and `CPT N` over memorizing r and t formulas

4.3 Keystrokes

- FV: $2,000, 8%, 5 yrs → `5 N` `8 I/Y` `0 PMT` `2000 +/− PV` `CPT FV` = **2,938.66**
- PV: $10,000 in 6 yrs, 7% → `6 N` `7 I/Y` `0 PMT` `10000 FV` `CPT PV` = **−6,663.42**
- r: $5,000 → $10,000 in 9 yrs → `9 N` `5000 +/− PV` `0 PMT` `10000 FV` `CPT I/Y` = **8.01%**
- t: $1,000 → $2,500 at 9% → `9 I/Y` `1000 +/− PV` `0 PMT` `2500 FV` `CPT N` = **10.63 yrs**
- Practice (a) FV of $8,000, 6.5%, 10 yrs → 15,017.10
- Practice (b) PV of $20,000 in 10 yrs at 6.5% → −10,654.52
- Practice (c) $8,000 → $20,000 in 10 yrs, r → 9.60%
- Practice (d) $8,000 → $20,000 at 6.5%, t → 14.55 yrs
- Trap: same sign on PV and FV → Error 5

Ch.5 Discounted Cash Flow Valuation

5.1 Multiple Cash Flows

- Discount / compound each cash flow at its own t, then **add** (never average); including CF0 → NPV
- FV: $4,000 now + $7,000 in 1 yr, 8%, value at year 3
  - `3 N` `8 I/Y` `0 PMT` `4000 +/− PV` `CPT FV` = 5,038.85
  - `2 N` `7000 +/− PV` `CPT FV` = 8,164.80 → sum = **13,203.65**
- PV of uneven CFs 200 / 400 / 600 / 800 at 12%
  - `CF` `2nd CLR WORK` `0 ENTER ↓` (CF0) `200 ENTER ↓↓` `400 ENTER ↓↓` `600 ENTER ↓↓` `800 ENTER ↓↓`
  - `NPV` `12 ENTER ↓` `CPT` = **1,432.93**

5.2 Annuity Types and Formulas

- Annuity: level CFs, finite (car / home loans) | Annuity due: paid at start (rent, lease, insurance) → × (1 + r)
- Perpetuity: C ÷ r (preferred stock, consol) | Growing perpetuity: $C_1 \div (r - g)$, r > g
- $PV = C \times [1 - 1/(1 + r)^t] \div r$ | $FV = C \times [(1 + r)^t - 1] \div r$
- r = rate **per payment period** (not APR); t = **number of payments** (not years)

5.3 Annuity Keystrokes

- PV (loan amount): $632 / mo, 1% / mo, 48 mo → `48 N` `1 I/Y` `632 +/− PMT` `0 FV` `CPT PV` = **23,999.54**
- Practice: $450, 0.75%, 36 mo → 14,151.06
- PMT: $20,000, 0.5% / mo, 60 mo → `60 N` `0.5 I/Y` `20000 PV` `0 FV` `CPT PMT` = **−386.66**
- Practice: $15,000, APR 9% monthly, 4 yrs → r = 0.75%, N = 48 → −373.28
- r: cost $1,000, $200 / yr × 8 → `8 N` `1000 +/− PV` `200 PMT` `0 FV` `CPT I/Y` = **11.81%**
- Practice: $8,000, $1,200 / yr × 12 → 10.45%
- N: save $2,000 / yr at 7% to reach $50,000 → `7 I/Y` `2000 +/− PMT` `0 PV` `50000 FV` `CPT N` = **14.95 yrs**
- Practice: $400 / mo, 0.5% / mo, $30,000 → 63.85 mo (5.32 yrs)
- FV: $300 / mo, 0.75% / mo, 30 yrs → `360 N` `0.75 I/Y` `300 +/− PMT` `0 PV` `CPT FV` = **549,223.04**
- Practice: $2,500 / yr, 7%, 25 yrs → 158,122.59

5.4 Annuity Due

- Method 1: ordinary value × (1 + r) | Method 2: `2nd BGN` `2nd SET` `2nd QUIT`, enter as usual, then **restore END**
- Ex: $1,000 / mo, 1% / mo, 36 mo → ordinary PV 30,107.51 × 1.01 = **30,408.58**
- Practice: $1,800, 0.5%, 24 mo → 40,613.16 × 1.005 = 40,816.22
- Lease PMT (BGN): $48,000, 12% APR monthly, 4 yrs → `48 N` `1 I/Y` `48000 +/− PV` `0 FV` `CPT PMT` = **1,251.51** (= ordinary 1,264.02 ÷ 1.01)

5.5 Perpetuities

- PV = C ÷ r | Ex: preferred $1.50 / qtr at 2% / qtr → **75.00**
- Growing: $C_1 \div (r - g)$ | Ex: $2.00 next year, g 3%, r 10% → 2.00 ÷ 0.07 = **28.57**
- If $2.00 was just paid ($D_0$): $C_1 = D_0(1 + g)$ = 2.06 → 2.06 ÷ 0.07 = **29.43**

5.6 APR vs. EAR

- APR = quoted rate (periodic rate × m) | EAR = actual annual rate | **always compare in EAR**
- EAR = $(1 + APR/m)^m - 1$ | APR = $m \times [(1 + EAR)^{1/m} - 1]$ | continuous: EAR = $e^r - 1$
- More frequent compounding / higher rate → larger APR–EAR gap
- `2nd ICONV`: NOM → C/Y → `CPT EFF`; reverse: EFF → C/Y → `CPT NOM` (standard BA II Plus operation)
- Ex: 15% APR → annual 15.00%, semiannual 15.56%, quarterly 15.87%, monthly 16.08%, daily 16.18%
- Practice: 9% APR → quarterly 9.31%, monthly 9.38%, daily 9.42%
- Bank A 15% daily (EAR 16.18%) vs Bank B 15.5% annual (15.50%) → A for deposits
- Card A 22.9% monthly (EAR 25.46%) vs Card B 23.5% annual (23.50%) → B is cheaper
- EAR 12% as monthly APR → **11.39%** | EAR 9% as quarterly APR → **8.71%**
- Calculator returns a monthly rate: $250 / mo × 15 yrs → $75,000 → `180 N` `250 +/− PMT` `0 PV` `75000 FV` `CPT I/Y` = 0.5303% / mo → APR **6.36%**, EAR **6.55%**

5.7 Loan Types and Amortization

- Pure discount (T-bill, zero) | interest-only (corporate bond) | amortized (home / car loan)
- Loan balance = PV of remaining payments
- Payment constant; interest falls, principal rises
- Interest = beginning balance × r | principal = PMT − interest | ending = beginning − principal
- Ex: $5,000, 9%, 5 yrs → `5 N` `9 I/Y` `5000 PV` `0 FV` `CPT PMT` = **−1,285.46**
  - Year 1: interest 450.00, principal 835.46, ending balance 4,164.54
  - Total paid 6,427.30 = principal 5,000 + interest 1,427.30
- Practice: $10,000, 8%, 4 yrs → PMT 3,019.21 | yr-1 ending balance 7,780.79 | yr-2 interest 622.46 | balance after 2 yrs 5,384.05

Ch.6 Interest Rates and Bond Valuation

6.1 Bond Anatomy and Keys

- Face / par = repaid at maturity | coupon = fixed interest | coupon rate = fixed at issue | maturity
- YTM = market required return, changes daily | after issue only the **price** changes
- Price = PV(coupon annuity) + PV(face)
- Keys: N = periods to maturity | I/Y = YTM per period | PMT = coupon per period | FV = face | PV = price
- PMT and FV positive → PV comes out negative (cash you pay); solving YTM → enter PV as negative

6.2 Par, Discount, Premium

- Coupon rate = YTM → **par**
- Coupon rate < YTM → **discount** (capital gain at maturity)
- Coupon rate > YTM → **premium** (capital loss at maturity)

6.3 Pricing Keystrokes

- Discount: $1,000, 8% coupon, 10 yrs, YTM 10% → `10 N` `10 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−877.11**
- Premium: same bond, YTM 6% → `10 N` `6 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−1,147.20**
- Semiannual: 9% coupon, 7 yrs, YTM 8% → N = 2 × yrs, I/Y = YTM ÷ 2, PMT = coupon ÷ 2
  - `14 N` `4 I/Y` `45 PMT` `1000 FV` `CPT PV` = **−1,052.82**
- Zero coupon: 5 yrs, YTM 6% semiannual → `10 N` `3 I/Y` `0 PMT` `1000 FV` `CPT PV` = **−744.09**

6.4 Yield to Maturity

- YTM = single discount rate that sets PV of promised CFs = price (the bond's IRR)
- Assumptions: hold to maturity | no default | coupons reinvested at YTM
- Annual: 8% coupon, 8 yrs, price 918.50 → `8 N` `918.50 +/− PV` `80 PMT` `1000 FV` `CPT I/Y` = **9.50%**
- Semiannual: 7% coupon, 10 yrs, price 900.29 → `20 N` `900.29 +/− PV` `35 PMT` `1000 FV` `CPT I/Y` = 4.25 → × 2 = **8.50%** (EAY 8.68%)
- Three yields (8% coupon, price 918.50): coupon rate 8.00% < current yield 80 ÷ 918.50 = 8.71% < YTM 9.50%

6.5 Interest Rate Risk

- Price and yield move **inversely** (promised CFs are fixed); price curve is convex
- Longer maturity and lower coupon → more sensitive; zero coupon is most sensitive
- Interest rate risk = price risk (not default risk)
- 30-yr bond, yield 10% → 12%: coupon 5% → −17.50% | 10% → −16.11% | 15% → −15.61%

6.6 Clean vs. Dirty Price

- Dirty (invoice) = clean (quoted) + accrued interest
- Accrued = coupon × (elapsed ÷ coupon period)
- Ex: semiannual $40 coupon, 2 months elapsed, quote $985 → 40 × 2/6 = 13.33 → dirty **998.33**

6.7 Fisher Effect and Yield Components

- **1 + R = (1 + r)(1 + h)** | Ex: nominal 15.5%, inflation 5% → 1.155 ÷ 1.05 − 1 = **10.00%** real (R − h = 10.5% is only approximate)
- Nominal yield = real rate + inflation premium + interest-rate-risk premium (rises with maturity) + bond-specific: default, taxability, liquidity premiums
- Higher expected inflation → higher nominal yields, lower bond prices

6.8 Term Structure

- Relation between maturity and yield (pure-discount Treasuries); yield curve = its graph
- Upward sloping → expected inflation up | downward → expected inflation down

6.9 Treasuries and Municipals

- Treasury: no default risk, state / local tax-exempt, federally taxable
- Municipal: default risk, federally tax-exempt → lower yield, held by high-bracket investors
- Tax-equivalent yield = muni yield ÷ (1 − t) | 6 ÷ 0.65 = **9.23%** (t = 35%) | 6 ÷ 0.80 = **7.50%** (t = 20%)

6.10 Debt vs. Equity

- Creditors: no ownership, no vote | interest is a pre-tax (deductible) expense | dividends are paid after tax
- Unpaid interest → default; unpaid dividend → not default
- Liquidation: equity is the residual claim (last)

6.11 Indenture, Call, Covenants

- Indenture = contract between issuer and bondholders; trustee represents holders
- Terms: face, coupon, maturity | security | repayment (sinking fund) | seniority | call | covenants
- Call: issuer may repurchase before maturity | call premium = call price − face | deferred call = protected period
  - Exercised when rates fall → bond value lower; investors demand a higher yield
- Negative covenants: limit dividends, pledging assets, mergers, asset sales, extra long-term debt
- Positive covenants: maintain NWC, supply audited statements, maintain collateral, keep ratios
- Firms accept covenants because the yield reduction exceeds the flexibility cost

6.12 Ratings and Bond Types

- Investment grade AAA / Aaa to BBB / Baa | junk (speculative) BB / Ba and below
- BBB− / Baa3 = last investment grade (institutional-investment cutoff)
- Zero coupon: only CF at maturity | Floating rate: collar = floor + cap | Convertible: lower coupon
- Put provision: opposite of call, adds value | CoCo | Sukuk: asset-based return instead of interest

6.13 Bond Market

- Large volume but mostly **OTC** (no central exchange); dealer quotes; thin liquidity per issue; institution-dominated
- Transparency problem → TRACE (2002) reporting and disclosure

Ch.7 Equity Markets and Stock Valuation

7.1 Stock vs. Bond Valuation

- Dividends at board discretion (coupons are contractual)
- No maturity → infinite cash flows
- Must estimate R and g; price is highly sensitive to both

7.2 Dividend Discount Model

- P_0 = PV of all future dividends
- "Sell later at a higher price" is already inside the model
- Non-dividend firms still have value only if they eventually pay
- No slide keystrokes for Ch.7 → solve with formulas

7.3 Four Growth Cases

- **Zero growth** (preferred): $P_0 = D \div R$
- **Constant growth**: $D_1 = D_0(1 + g)$; $P_0 = D_1 \div (R - g)$, requires $R > g$; $P_t = D_{t+1} \div (R - g)$
- **Nonconstant**: list $D_1 \ldots D_t$, find P_t, discount all; P_t is discounted t periods
- **Two-stage** (g_1 for t yrs, then g_2): $P_0 = D_1/(R - g_1) \times [1 - ((1 + g_1)/(1 + R))^t] + P_t/(1 + R)^t$, with $P_t = D_{t+1}/(R - g_2)$ (standard form; slide formula is an image)
- Sustainable growth is low single digits
- Zero: D \$3.00, R 10% → **30.00**
- Constant: D_0 \$2.00, g 5%, R 12% → D_1 = 2.10; 2.10 ÷ 0.07 = **30.00**
- Future price: $D_6 = 2.00 \times 1.05^6$; $D_6 \div 0.07$ = **38.29**
- Nonconstant: D_0 \$1.00, 20% growth for 3 yrs then 5% forever, R 15%
  - $D_{1\ldots3}$ = 1.20, 1.44, 1.728 | D_4 = 1.8144 | P_3 = 1.8144 ÷ 0.10 = 18.144
  - $P_0 = 1.20/1.15 + 1.44/1.15^2 + 1.728/1.15^3 + 18.144/1.15^3$ = **15.20**
- No dividend for 4 yrs, \$0.50 at end of yr 5, g 4%, R 20% → $P_4 = 0.50 \div (0.20 - 0.04) = 3.125$; $\div 1.20^4$ = **1.51**

7.4 Return Decomposition

- **R = dividend yield ($D_1 \div P_0$) + capital gains yield (= g under constant growth)**
- Dividend yield is like a bond's current yield
- Ex: P \$20, D_1 \$1, g 10% → 5% + 10% = **15%**

7.5 Multiples

- P/E, price-sales: fallback when there is no dividend
- Benchmark = comparable firms or own history; "comparable" drives the answer
- Losses → P/E meaningless; P/S still works
- Price = EPS × benchmark P/E | Ex: 3.00 × 18 = **54.00**
- Price = sales per share × benchmark P/S | Ex: 10.00 × 2.5 = **25.00**

7.6 Common Stock Rights

- Elect directors; vote (mergers, charter amendments, new shares)
- Dividends only after board declaration | residual claim | limited liability (loss up to paid-in)
- Preemptive right: first claim on new shares, prevents dilution
- Proxy: delegated vote (management usually favored); proxy fight: rival solicits proxies
- Dual-class: Class B 10 votes vs Class A 1 (founder control); one-share-one-vote varies by law / listing rules

7.7 Dividends

- Discretionary (non-payment ≠ default)
- Paid from after-tax income (asymmetric to deductible interest)
- US corporate dividends-received deduction (DRD) → corporations prefer preferred stock

7.8 Preferred Stock

- Fixed dividend on par; paid before common, after creditors
- Cumulative: arrearages paid first | normally no vote (may gain vote in arrears)
- Looks like debt but is not: missed dividend ≠ bankruptcy trigger; dividends not tax-deductible; equity legally and in accounting

7.9 Stock Markets

- Primary (firm receives funds) vs secondary (firm receives nothing)
- Dealer: holds inventory, bears risk | broker: agent, earns commission
- Bid = dealer buys | ask = dealer sells | spread = dealer's compensation
- NYSE: hybrid (DMM, mostly electronic) | NASDAQ: dealer market, competing market makers, ECNs narrow spreads
- Quote fields: Last / Close, Net change, Bid / Ask, Volume, Dividend, Yield = dividend ÷ price, P/E = price ÷ EPS

Exam Trap Checklist

T.1 Calculator

- Same sign on PV and FV → Error 5; cash out −, cash in +
- `2nd CLR TVM` / `2nd CLR WORK` before each problem; P/Y = 1
- Match units of N and I/Y (semiannual: N = yrs × 2, I/Y = rate ÷ 2)
- First CF now → annuity due = ordinary × (1 + r); turn BGN off afterwards

T.2 Statements and Ratios

- Interest deducted after EBIT | NCS adds depreciation | OCF adds depreciation back
- Inventory turnover uses COGS | quick ratio excludes inventory | EM = 1 + D/E | M/B uses book value per share
- DuPont is a product, not a sum

T.3 Rates and Bonds

- APR given → convert to the periodic rate first; EAR = $(1 + APR/m)^m - 1$; compare in EAR
- After-tax interest = interest × (1 − T); dividends paid from after-tax income
- Price and YTM move inversely; coupon rate < YTM → discount; clean + accrued = dirty
- Tax-equivalent yield: **divide** by (1 − t) (6% × 0.65 = 3.90% is wrong)

T.4 Stocks and Written Answers

- Constant growth numerator is D_1 ($D_0 \times (1 + g)$); valid only if R > g
- P_3 is discounted 3 periods, not 4; losses → use P/S
- Written answers: definition + direction (up / down) + reason in one line